ME 4702-A EV and the Grid
4/9/2026
Group 3: Hyungjun Park, Kevin Andersen, Andrew Schnupp, Alec Grosswald, Ryland Rembert
Professor Mike Tinskey

Module 2.1: Baseline Energy Measurement

Introduction

The aim of this module was to determine a baseline energy measurement for a single lap of the cart competition track in order to validate a method for recording cart energy activity during operation. A custom arduino script was designed to accurately pull cart data outputs during live operation, filing the logged data to a micro-sd card, which was then processed through a matlab script for sensible data analysis. To extend the scope of our understanding, data analysis was performed with and without an onboard driver through use of a remote controller, giving greater insight into cart limits, as well as enabling greater energy efficiency and quicker lap times.

Setup

In order to begin sampling data, we first had to interface with the carts internal current and voltage sensors. Using the given equipment, in combination with arduino code, the team determined the corresponding output of each pin within the system, marking each pin with its associated function (voltage, current, etc). With the pins marked, a custom arduino script was written to output readings from each pin in live time.

Data Logging

While the readings outputted by the arduino script could be visually observed on the associated LCD screen, there was no way to collect this data for later analysis without active connection to a computer. A driver could theoretically operate the cart with a laptop in their lap, but this was determined to truly simulate a baseline energy measurement for a single lap of the track. Initially, hardware was designed to connect onboard measurements to a phone app through bluetooth, but this method lacked a refined output, primarily due to lag between cart operation and data output on the phone.

To combine the benefits of the wired onboard collection/output (computer) and the practicality of the wireless app, focus shifted to using a micro sd card to log data onboard during operation, which could later be connected to a computer for data analysis. This approach allowed for precise data measurements to be collected without the need for a clunky setup of physically holding the output device. Data collection was achieved with true similarity to a race-like test environment, allowing future improvements and optimization to be accurately analyzed.

Power Output

Data from the micro sd card was imported into matlab for analysis. Battery voltage was plotted continuously during a lap, shown in Figure 1. The output is noisy during the run, but shows stability before and after the lap occurred, somewhat validating the results. More confirmation comes from the trends during the lap: the voltage drops quickly during the initial 30 seconds of drive time because this is the uphill section of the course, whereas in the downhill section the voltage gradually increases, evidence of potential regenerative braking effects.

Further evidence of regenerative braking comes from the power output plot, shown in Figure 2. Battery power was calculated by multiplying the battery current and voltage. Huge power draws occur at the start of the lap during the uphill section, and you can visually see points at which the driver stops accelerating to change direction (cart does not change direction smoothly). A bit after the 30 second mark, the driver rounds the first corner of the track, beginning the downhill section. After this point, we start seeing areas of negative power, highlighted in red, resulting from negative current readings. This ultimately confirmed the existence of regenerative braking, highlighted in green, lining up with the findings from the voltage output. To find the total power consumption for a lap (thereby satisfying the baseline energy measurement) we had to convert power from Watts to Watt-hours. To do so, we initially integrated the power function over time using both an arduino script and the "cumtrapz" function in matlab. However, due to the lack of granularity in the voltage readings, we found both these methods to be significantly inaccurate, as supported later by our model validation. The approximation method used by the code fails to accurately capture the spiky voltage reading caused by an inherently noisy sensor with a relatively low sampling rate. A far more accurate method for finding lap power consumption was to use the formula P [kWh] = (I [W] * V [V] / 1000) * (t [s] / 3600). Power consumed (P) was found by simply multiplying the current, voltage, and time at any moment (with some adjustment for units). By subtracting the max and min of these values throughout the lap, an accurate baseline energy measurement was taken.

Model Validation

To validate the power consumption findings, our calculated value was compared to the battery percentage loss shown in the carts associated bluetooth app. The app showed a drop from 60-46% battery over the course of 3 laps around the track, simplified to 4.67 % per lap. The Segway Ninebot Model S has a 236 Wh pack, where 4.67% of 236 Wh is 11.02 Wh consumed per lap. As previously mentioned, we originally calculated power consumption using two methods. The first (integrating the power output) resulted in a power loss of ~3 Wh. The second (subtracting the max and min of I*V*t) resulted in a power loss of 11.24 Wh. These results informed the choice of using the second method for accurate results.

Remote Vehicle Control

To extend the scope of our knowledge, we also ran the same analysis procedure for a test lap without a rider through the use of a remote controller. The resulting voltage readings are shown in figure 3, where it's visible that the maximum amplitude of the voltage readings is significantly reduced from when a driver was present. Figure 4 shows the power output, where again the amplitude is greatly reduced, and additionally there was much less regenerative braking present. The power outputs are directly compared in figure 5, where it's clear that the remote controlled lap consumes much less power.

Improvements & Optimization

Interestingly, we found that the remote controlled lap (without a person in the cart) was slower than the lap with the driver, shown in table 1. After further testing, we found this to be because the cart caps its own max speed. The added weight of the driver is therefore not punished on the uphill sections of the course, but ends up benefitting the speed of the run on the downhill section where the max speed of the cart can be exceeded through stored momentum. This does of course come at the cost of energy consumption, where the energy consumed using remote control was nearly half that of the rider consumption. Overall, the team came to understand how to optimize lap time by attempting to spend more

time sitting at max speed on the uphill, cornering sharply and quickly (since max speed is quickly achieved), and braking on downhill sections to fully take advantage of our stored gravitational energy.

Figures:

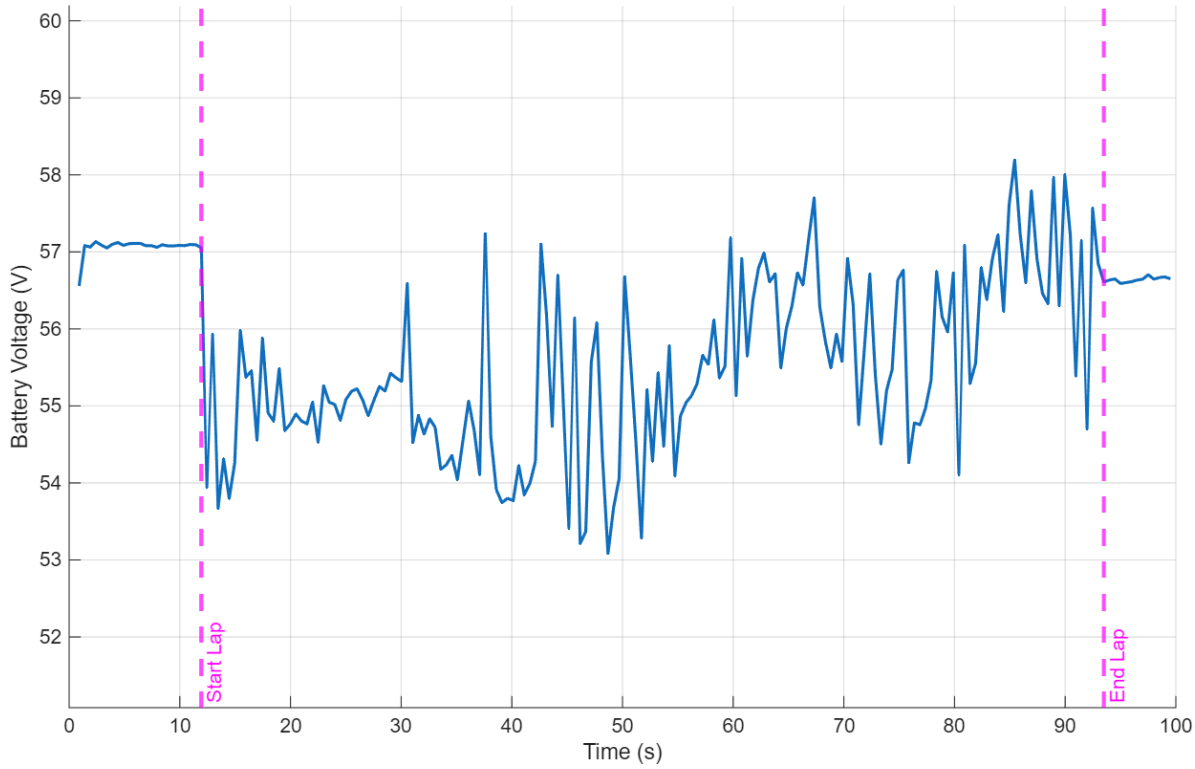

Figure 1: Battery Voltage (with Rider)

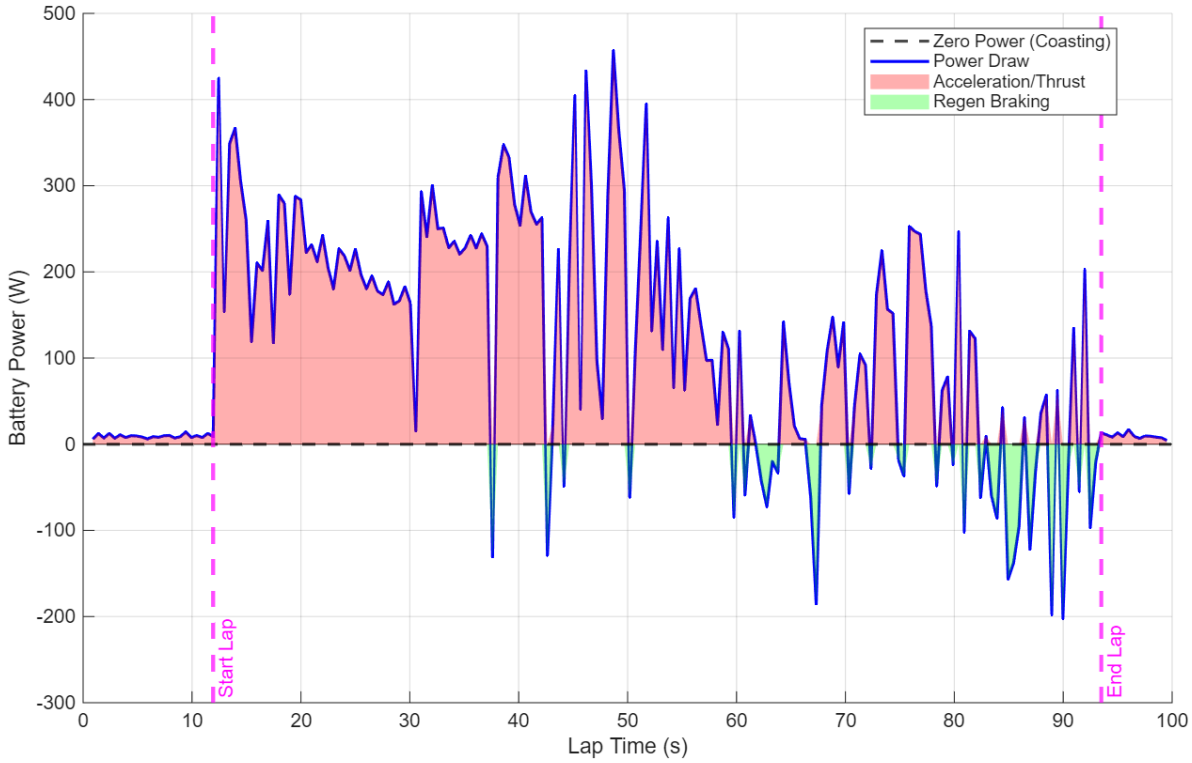

Figure 2: Scooter Power Output (with Rider)

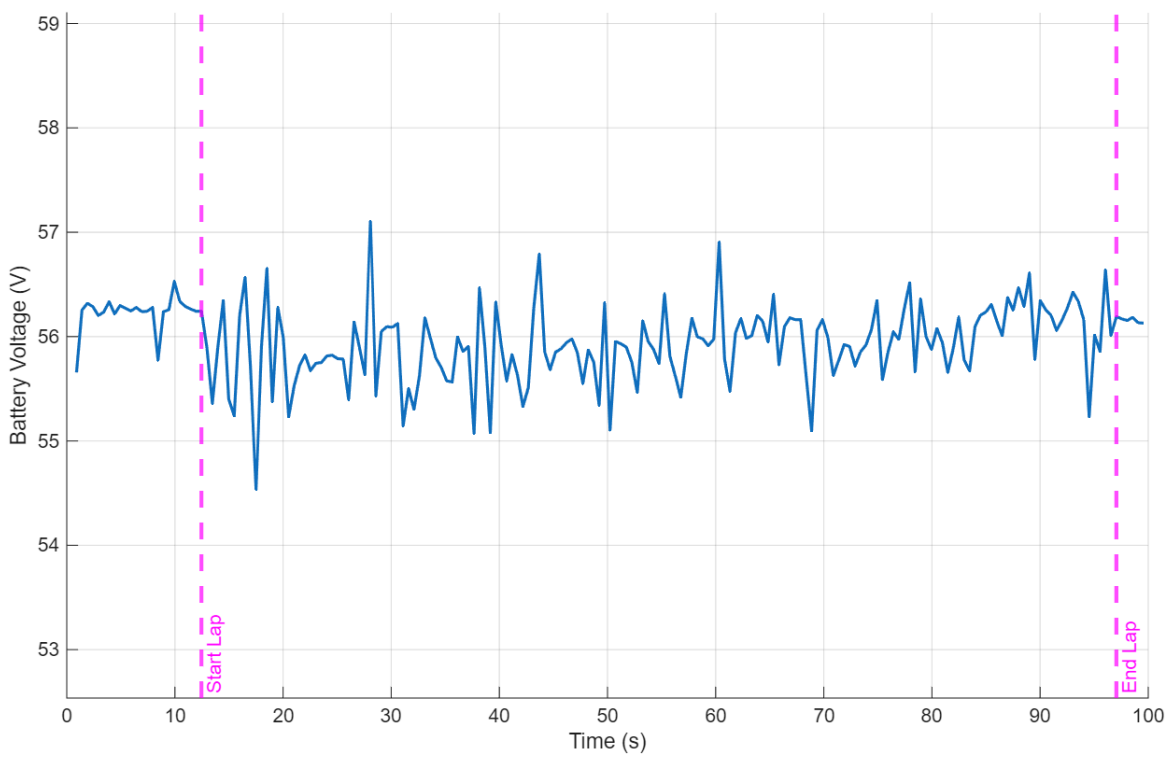

Figure 3: Battery Voltage (Remote Controlled)

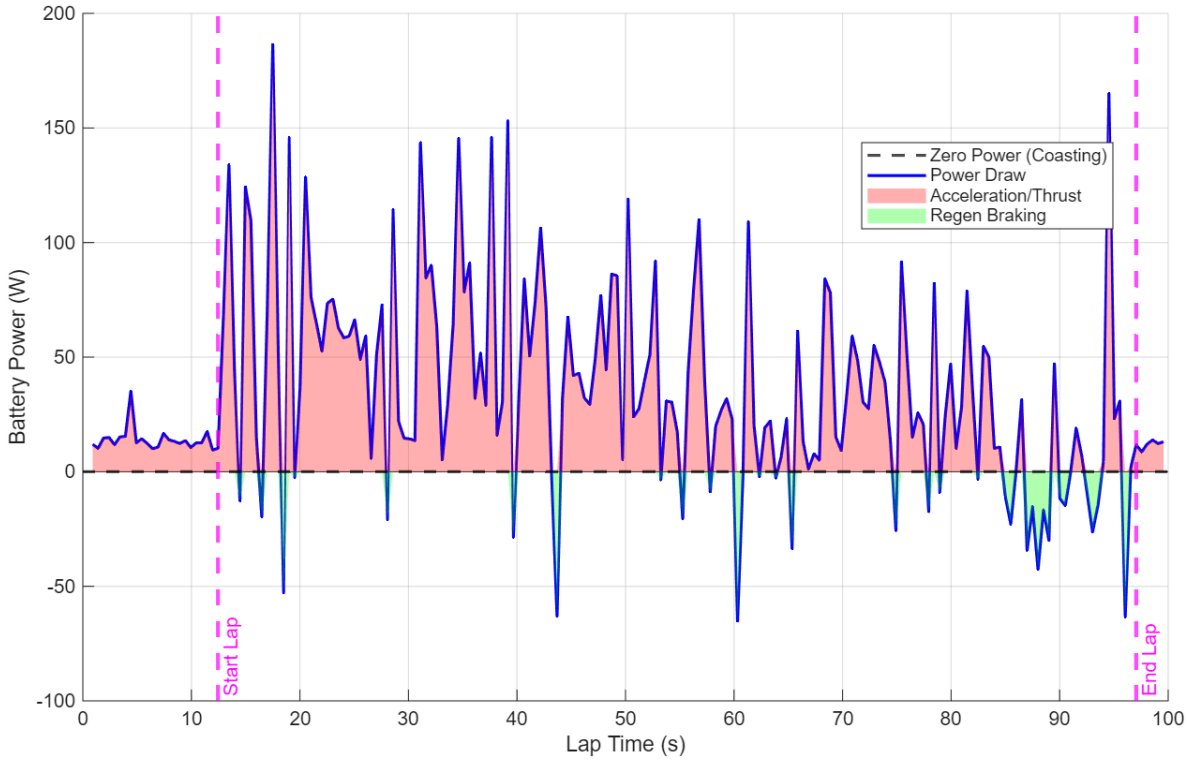

Figure 4: Scooter Power Output (Remote Controlled)

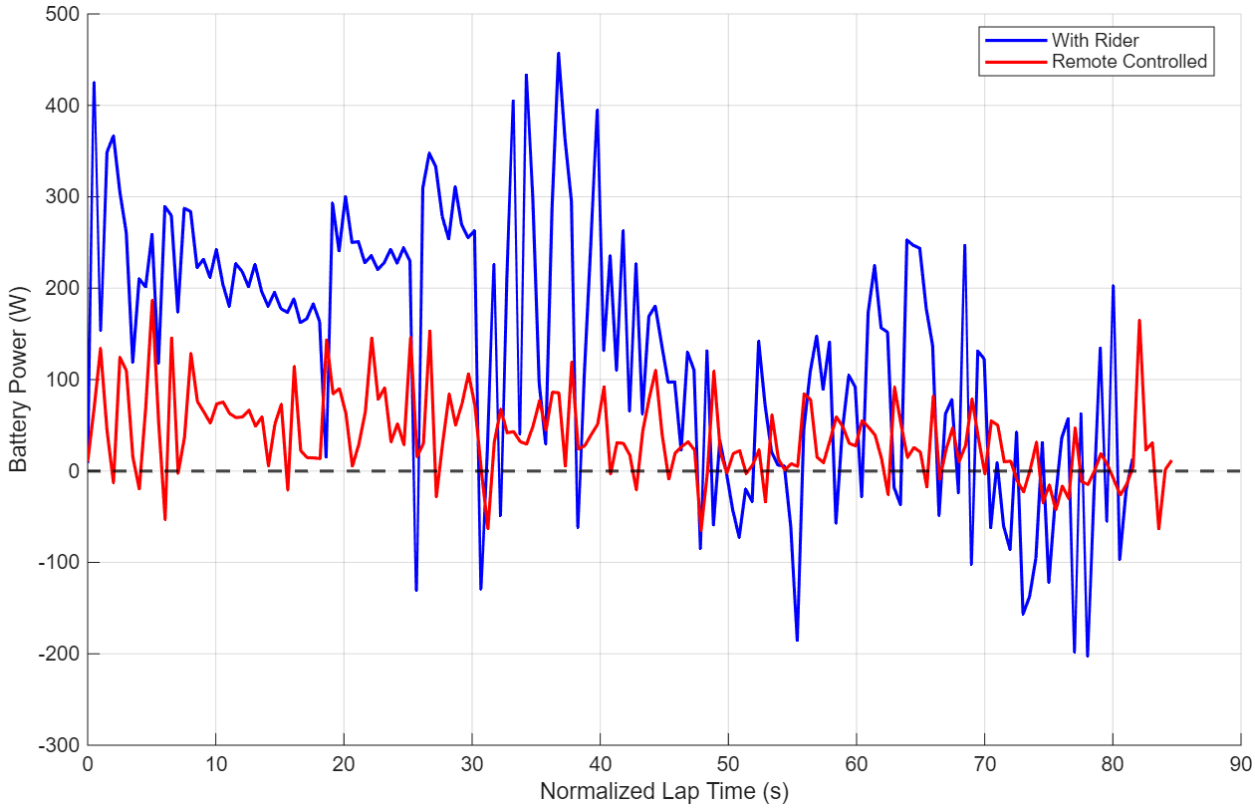

Figure 5: Power Output Comparison (Lap Only)

Tables:

Table 1 : Lap Comparison (Rider vs Remote Controller)

	With Rider	Remote Controlled
Lap time (s)	81.54	84.59
Battery Energy Used (Wh)	11.24	6.02
Peak Acceleration Power (W)	456.86	186.43
Peak Regen Braking Power (W)	-202.42	-65.19
Max Voltage Droop (V)	3.97	1.71